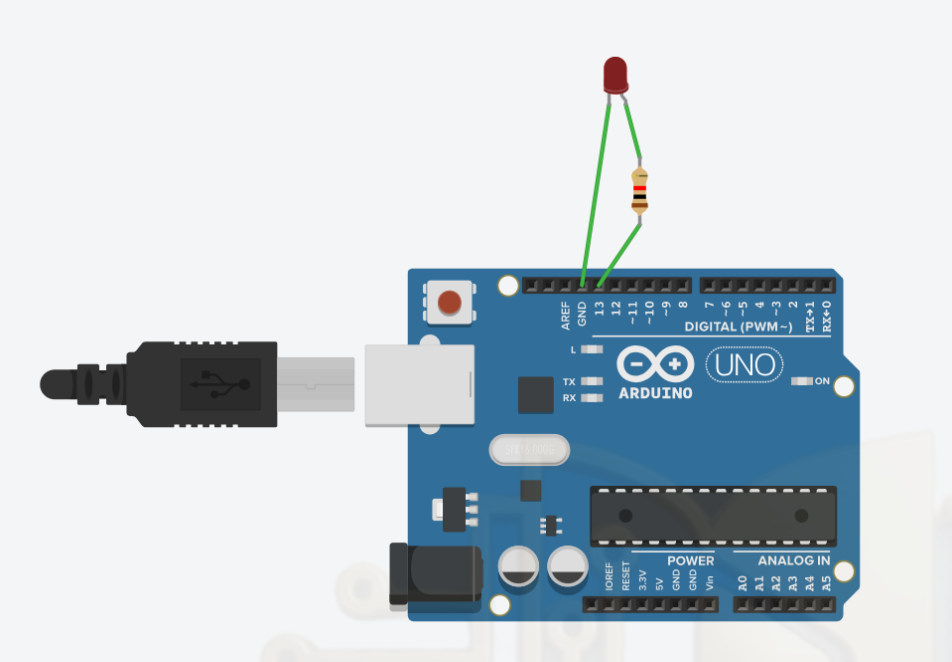
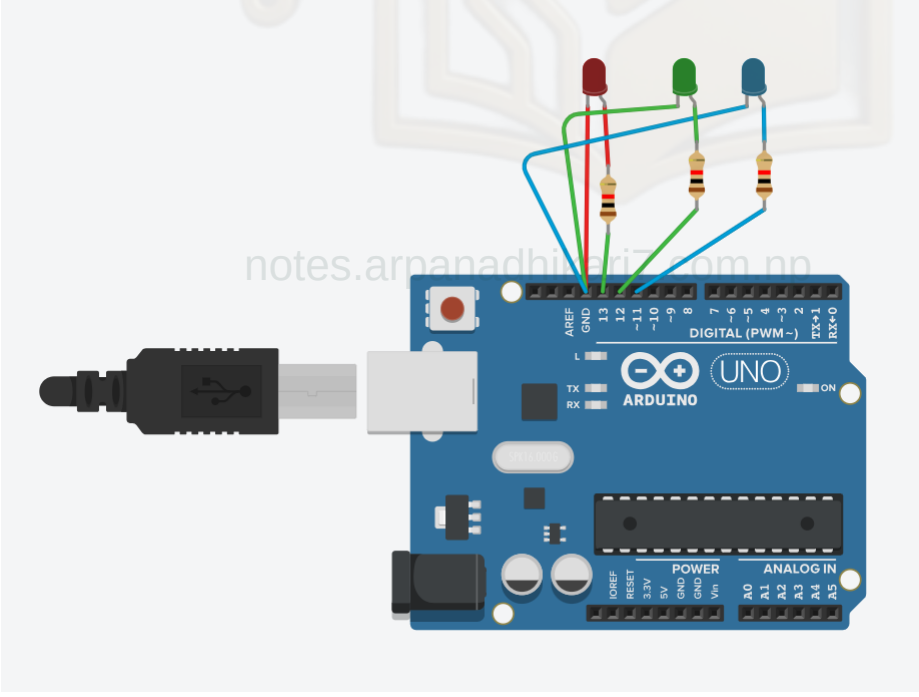
IOT Projects Using Arduino

1. **Simple LED on and Off**

Code Circuit Configuration

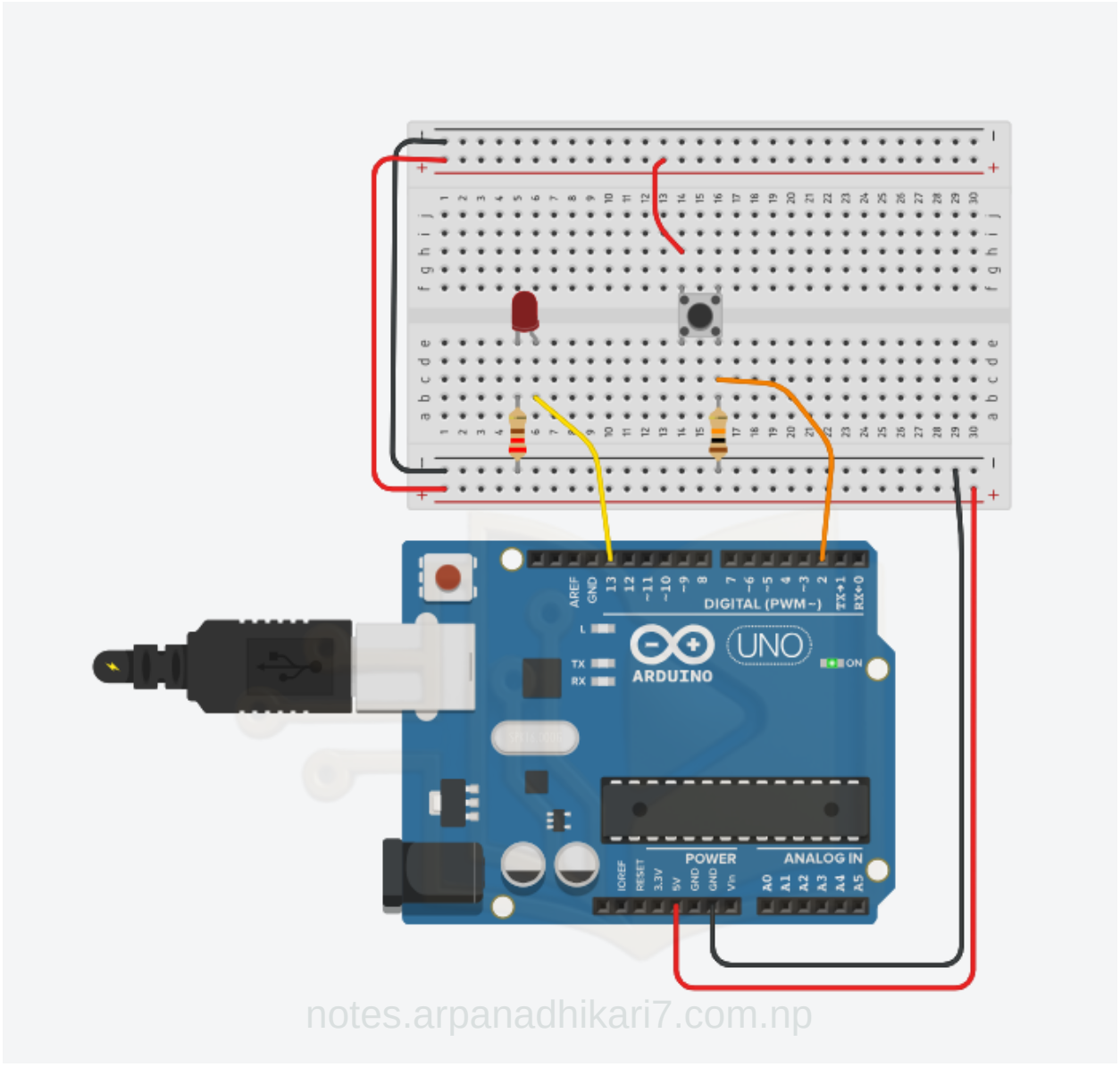


2. **3 led RGB with a delay of 2s**

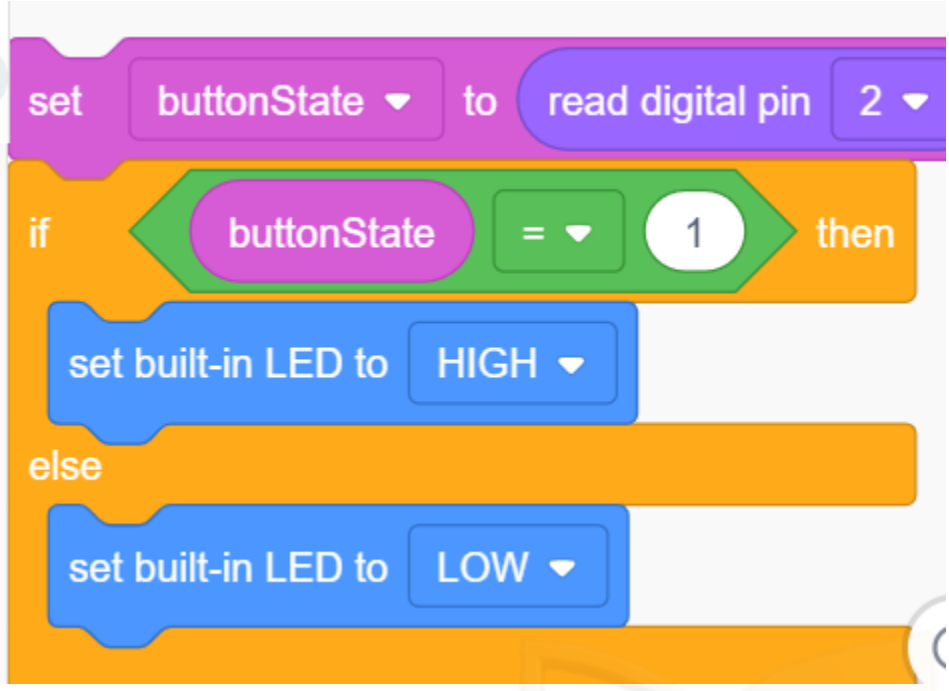


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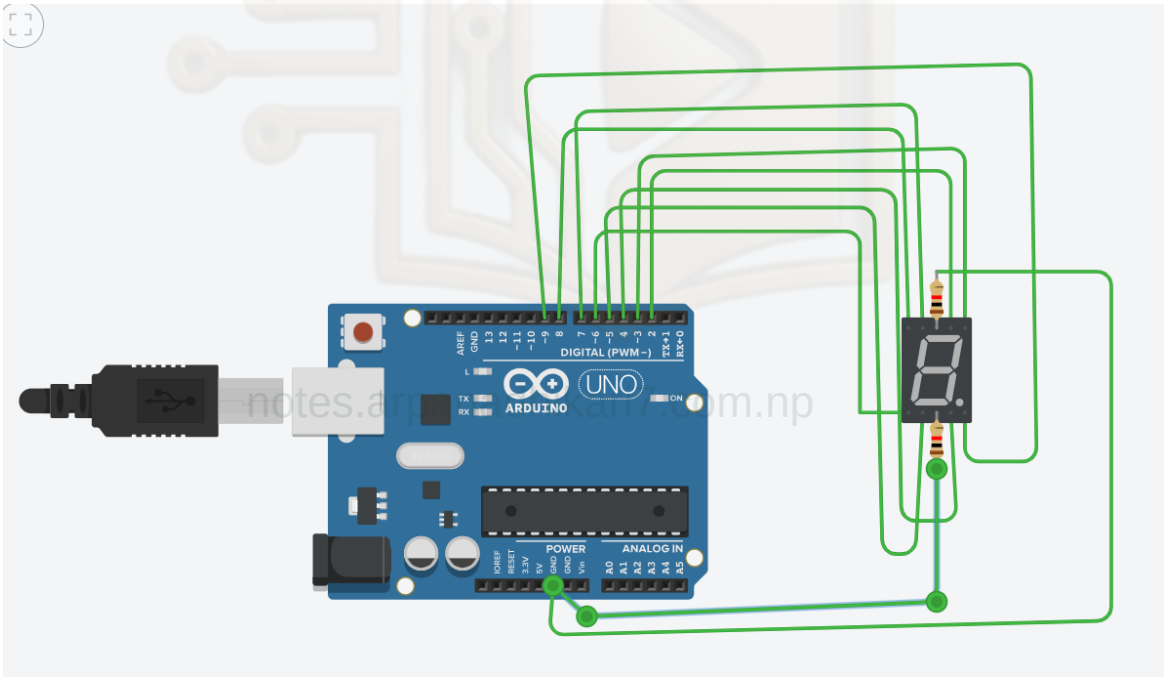
3. Led On and Off with push button



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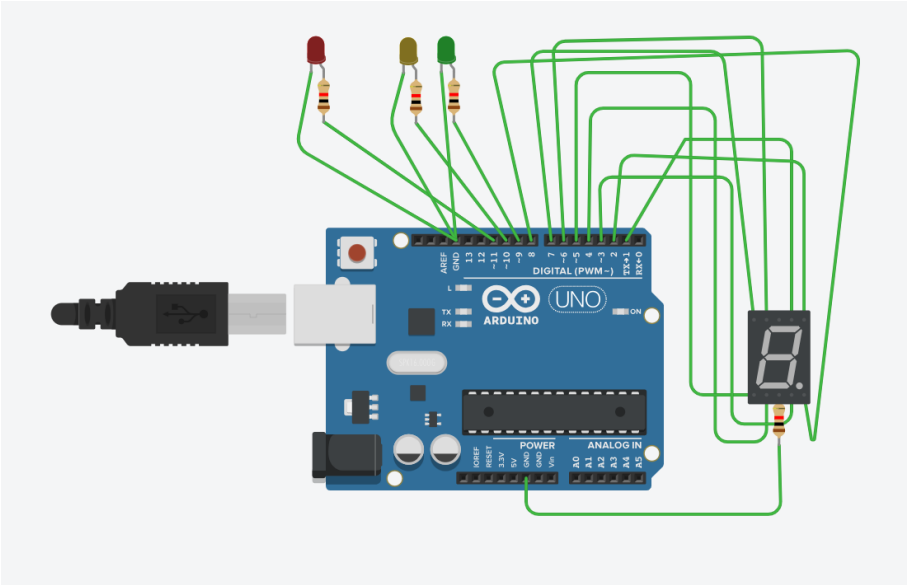


4. **7-Segment Display**



5. **7-segment display with LED traffic light**

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Program

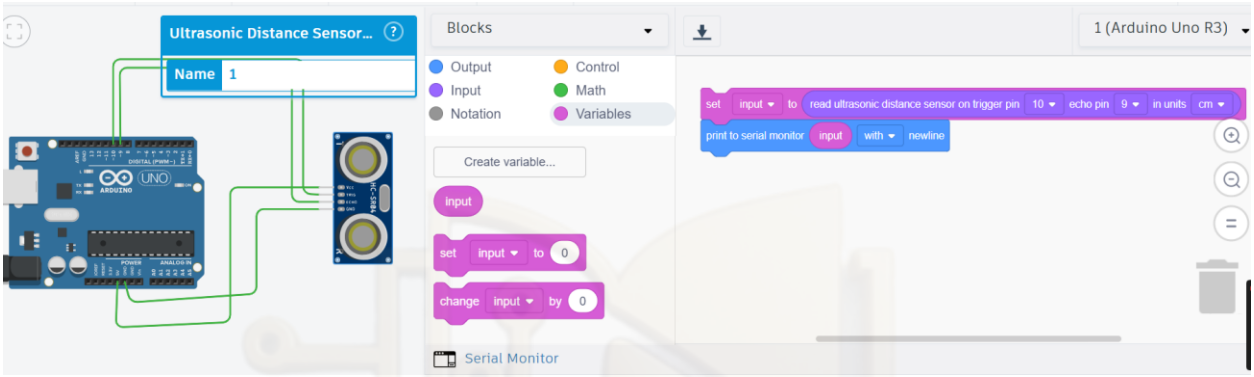
```
int count = 5;
// a b c d e f g
int segPins[7] = {1,2, 3, 4, 5, 6, 7};
// digits 0–5
byte digits[6][7] = {
  {1,1,1,1,1,1,0}, // 0
  {0,1,1,0,0,0,0}, // 1
  {1,1,0,1,1,0,1}, // 2
  {1,1,1,1,0,0,1}, // 3
  {0,1,1,0,0,1,1}, // 4
  {1,0,1,1,0,1,1}  // 5
};
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
  for (int i = 0; i < 7; i++) {
    pinMode(segPins[i], OUTPUT);
  }
}
void displayDigit(int num) {
  for (int i = 0; i < 7; i++) {
    digitalWrite(segPins[i], digits[num][i]);
  }
}
void loop() {
  displayDigit(count);
  digitalWrite(LED_BUILTIN, HIGH);
```

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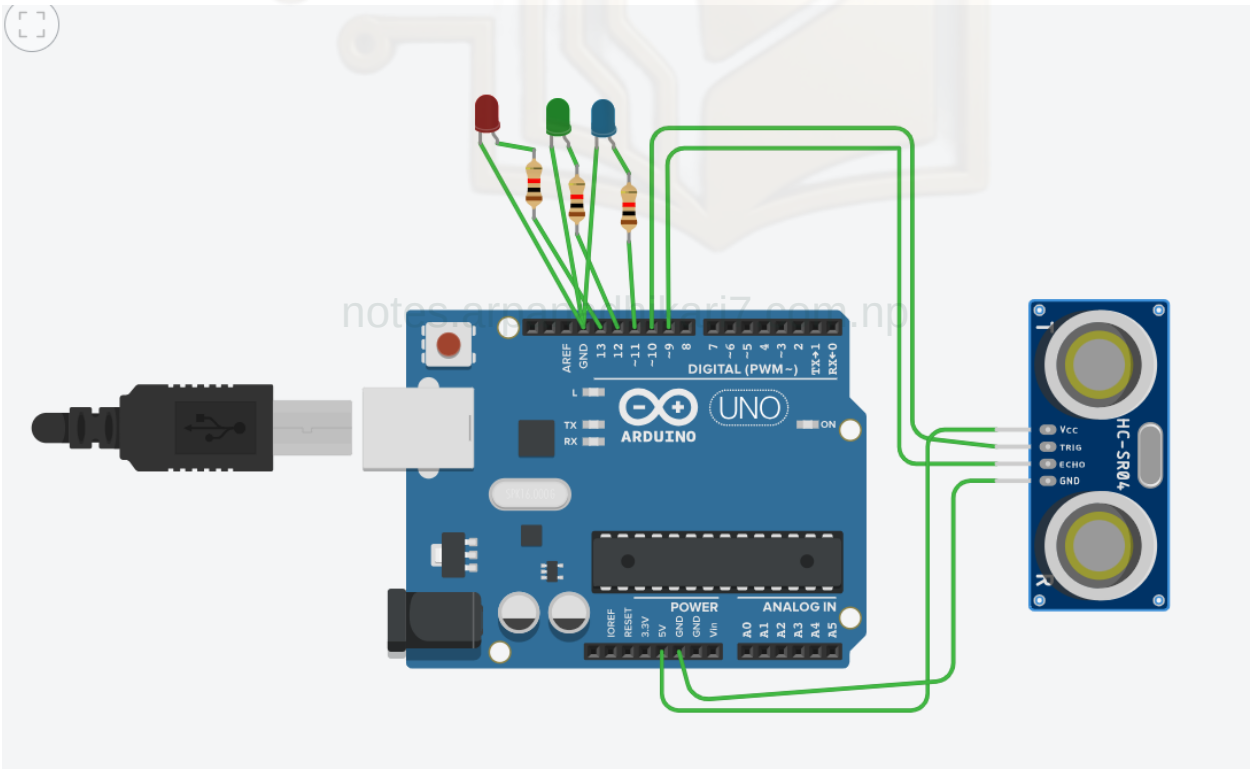
```
delay(5000);
digitalWrite(LED_BUILTIN, LOW);
delay(2000);
count--;
if (count < 0) {
  count = 5;
 }
}
```

6. **Ultrasonic Sensor Interfacing**

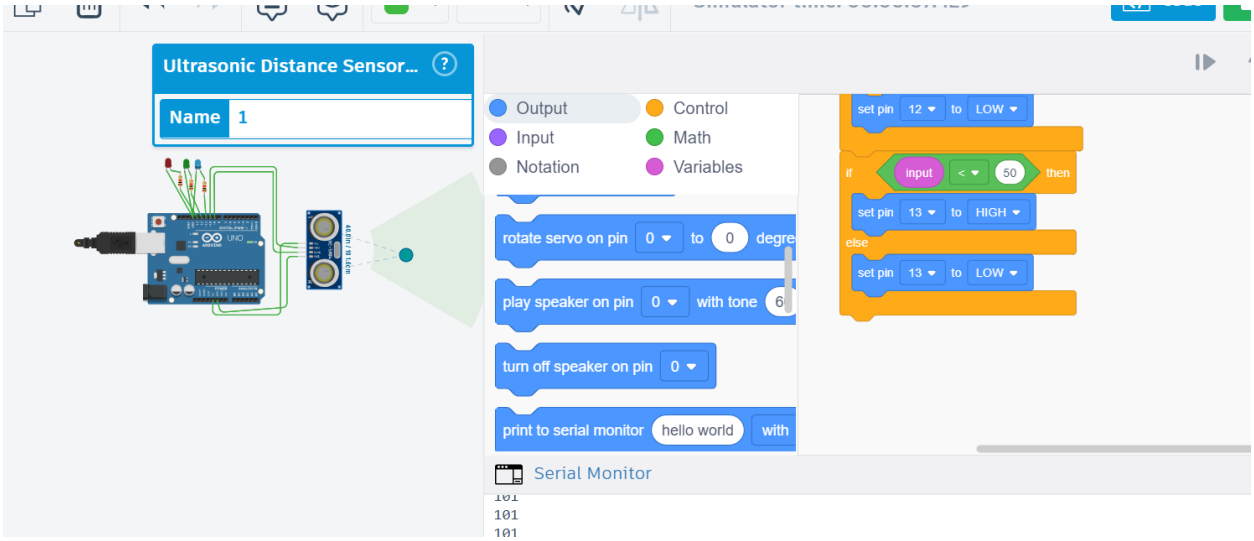
   Create an input variable and attach



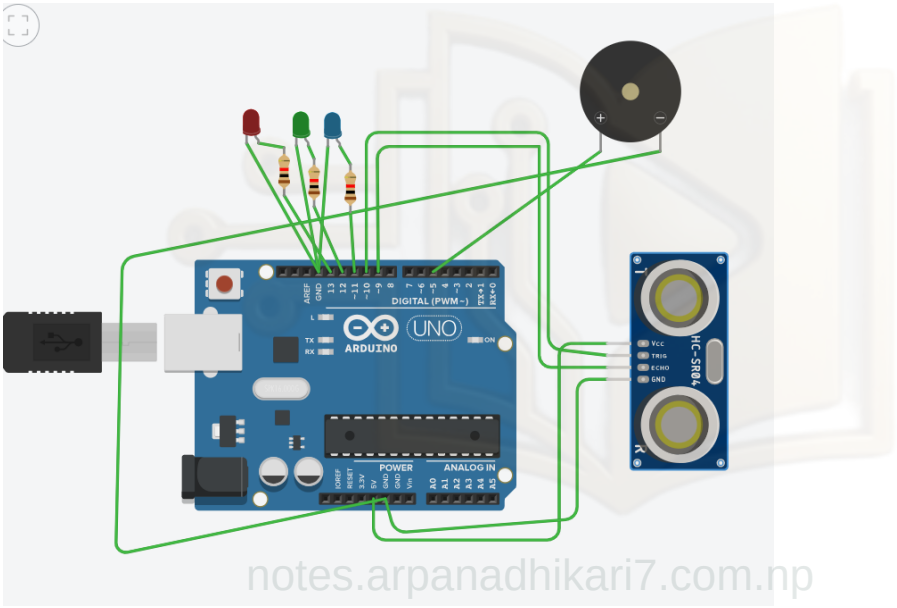
7. **Distance Measurement With LED & Arduino and Ultrasonic sensors**



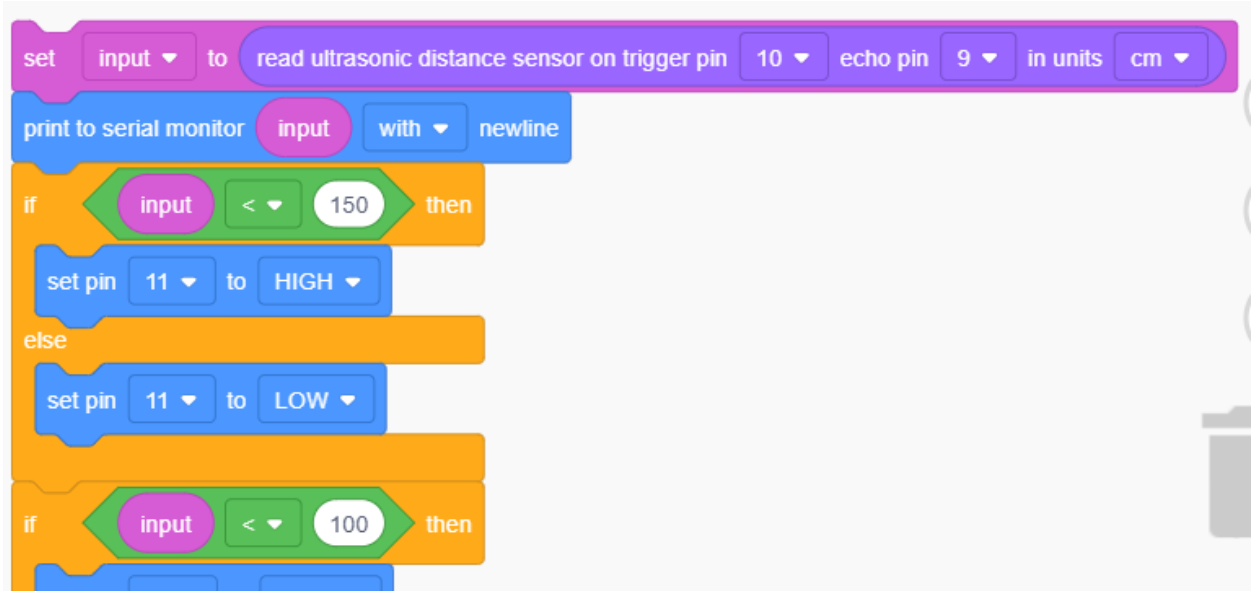
Available on notes.arpanadhikari7.com.np

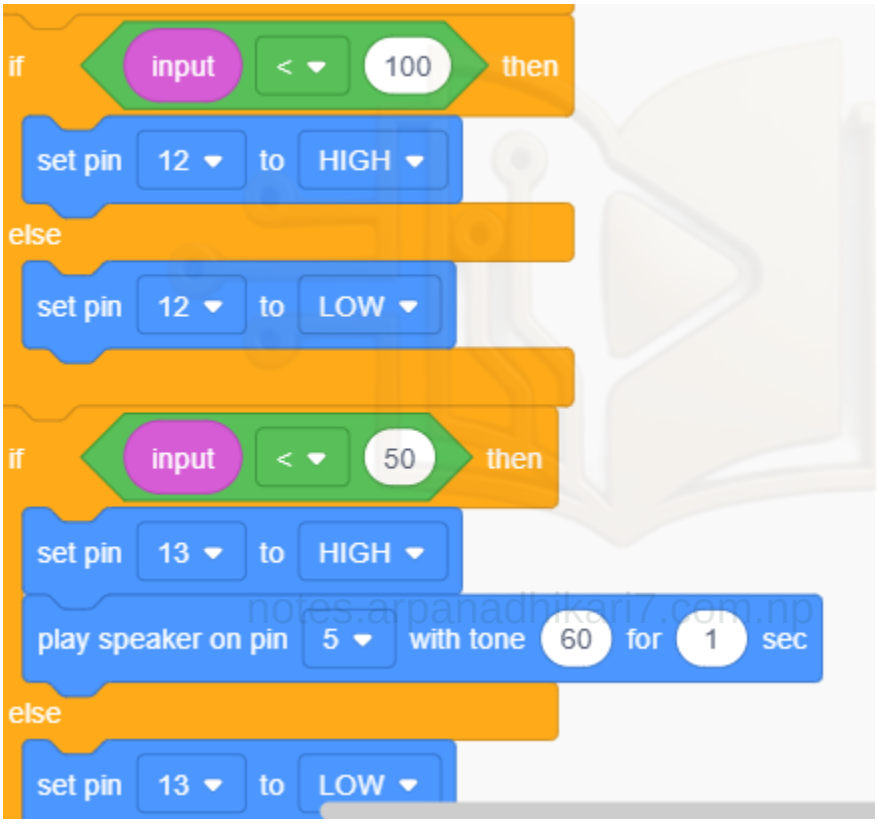


8. **With an alarm as well**



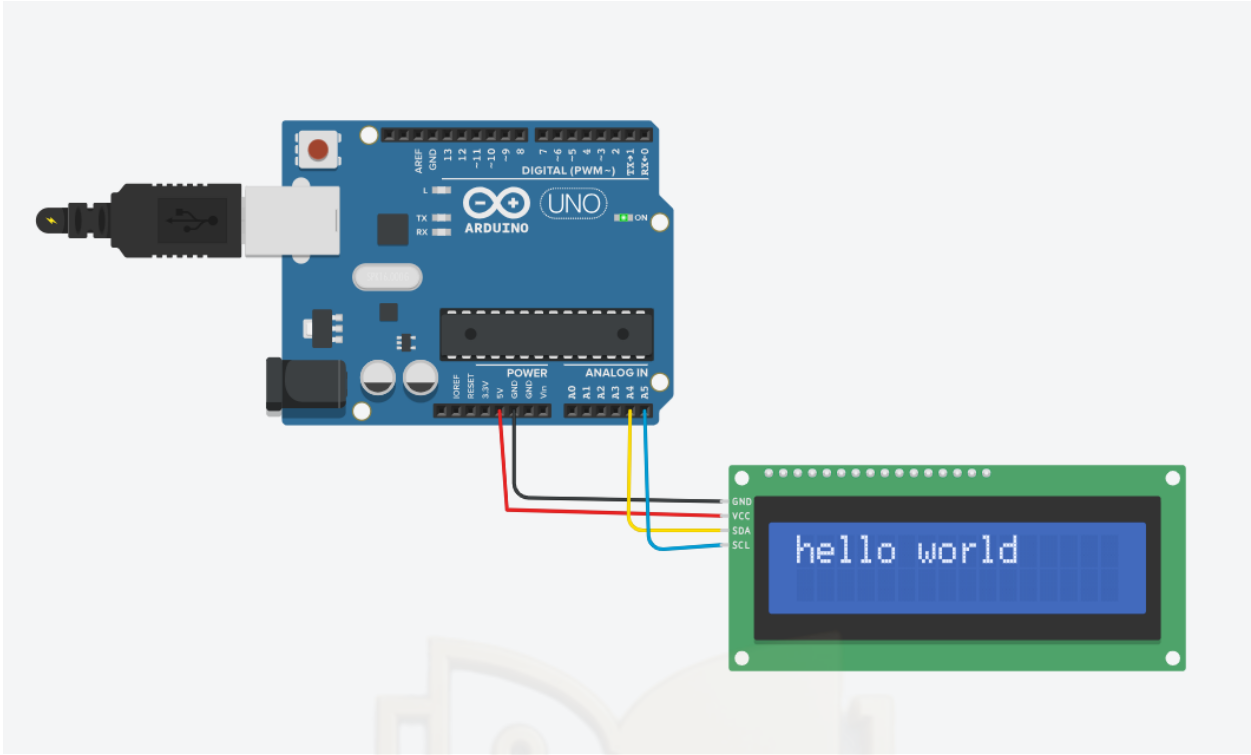
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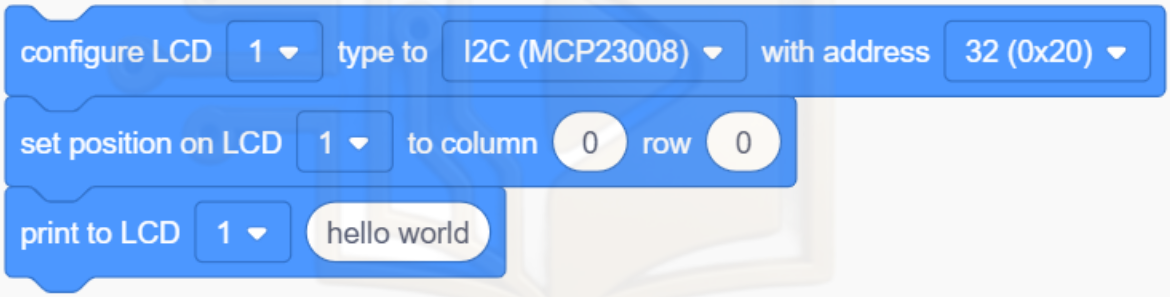




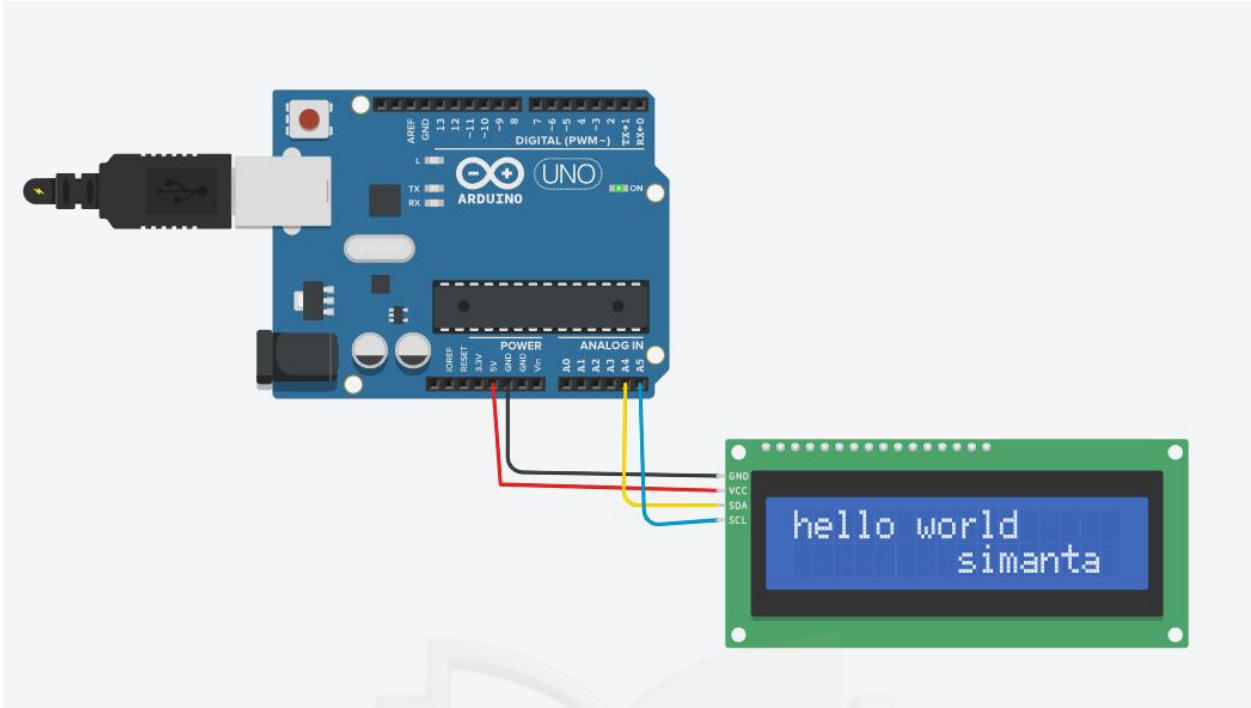
9. **Print Hello world**

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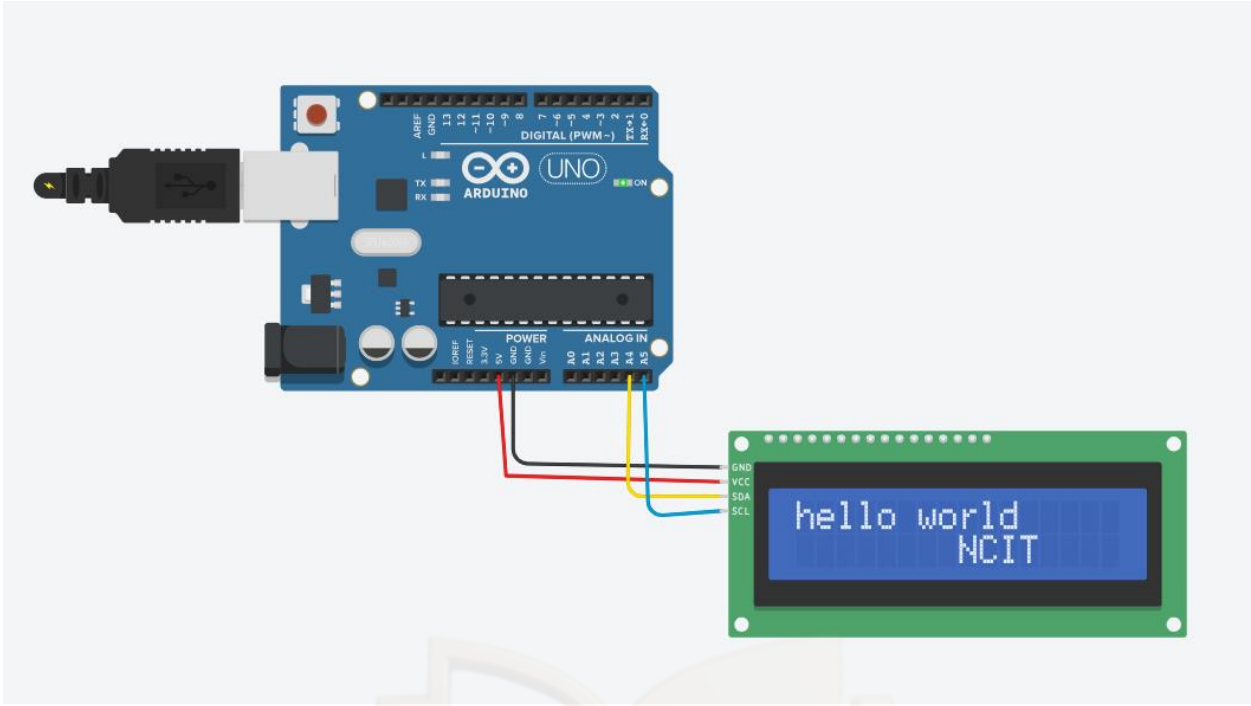


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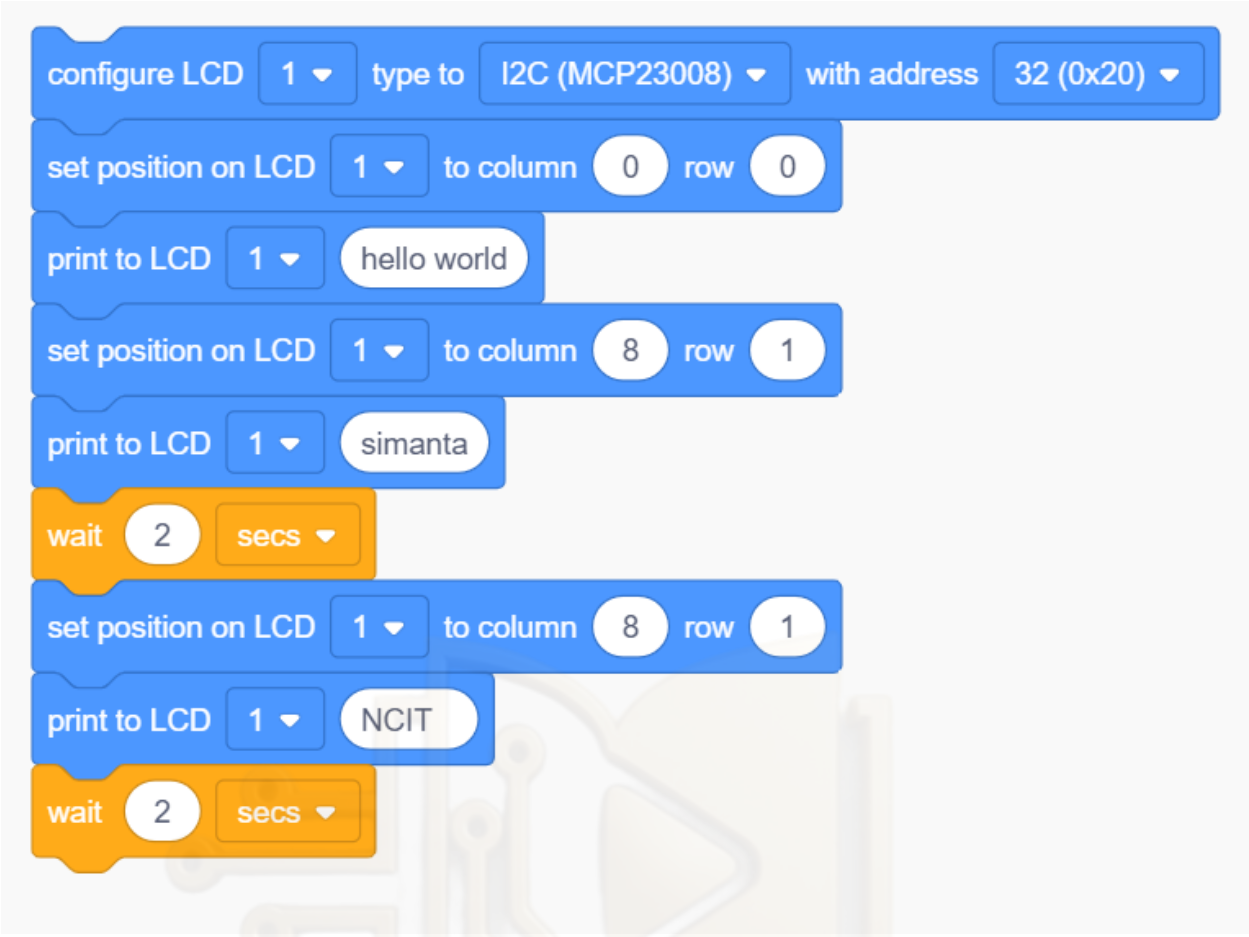




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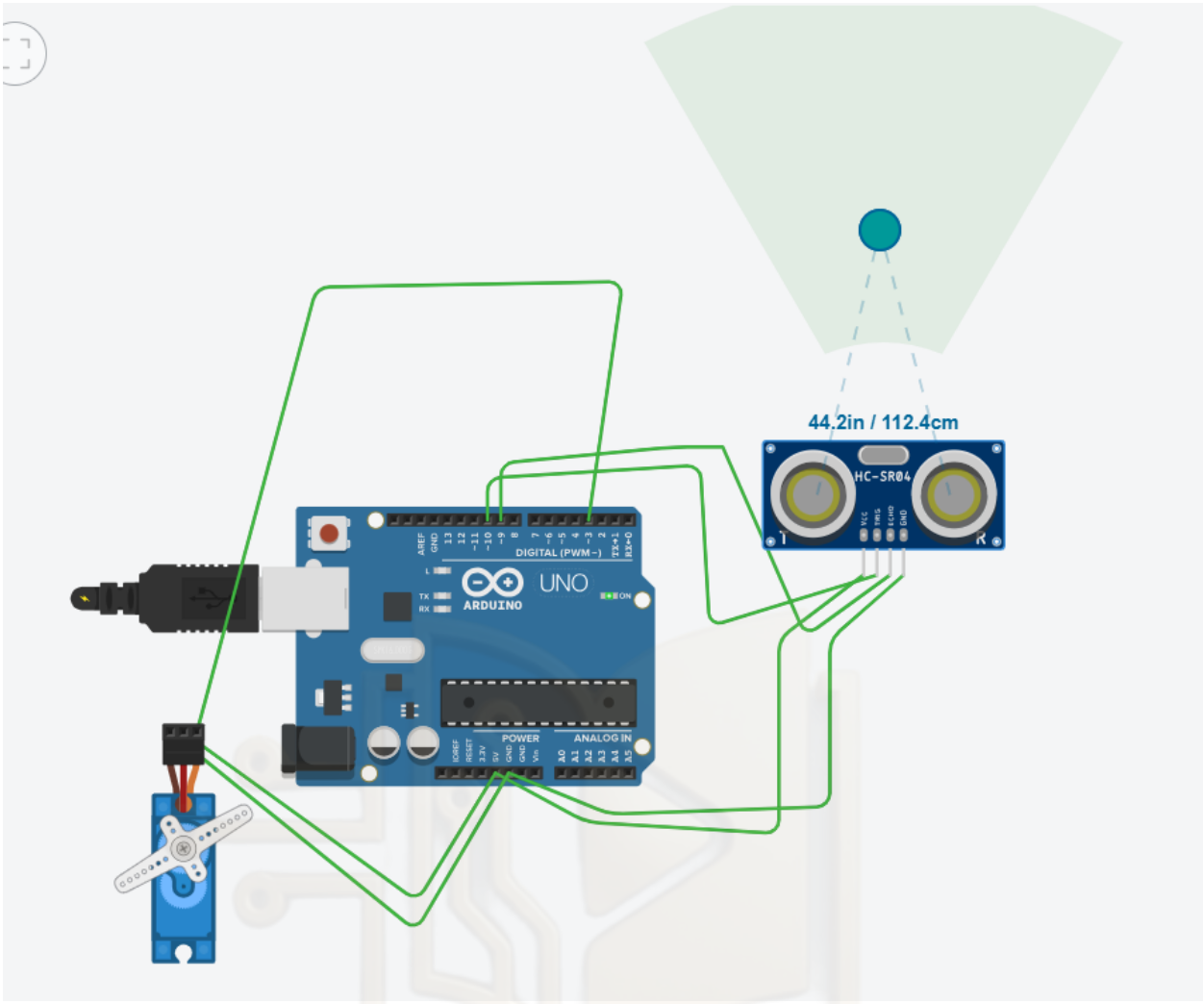


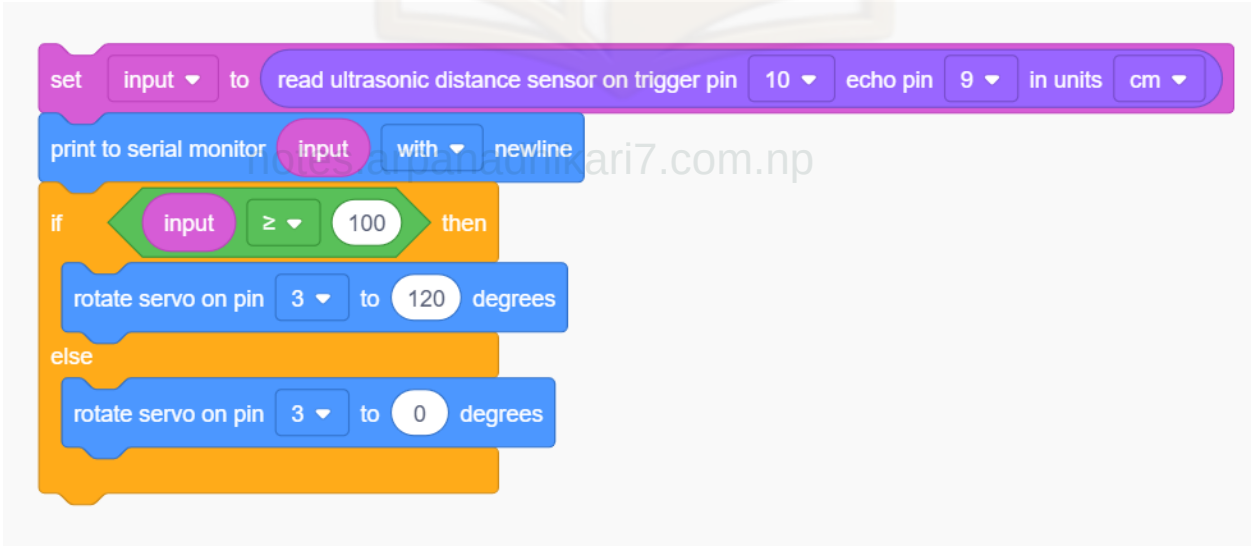
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10. Ultrasonic sensor with servo motor

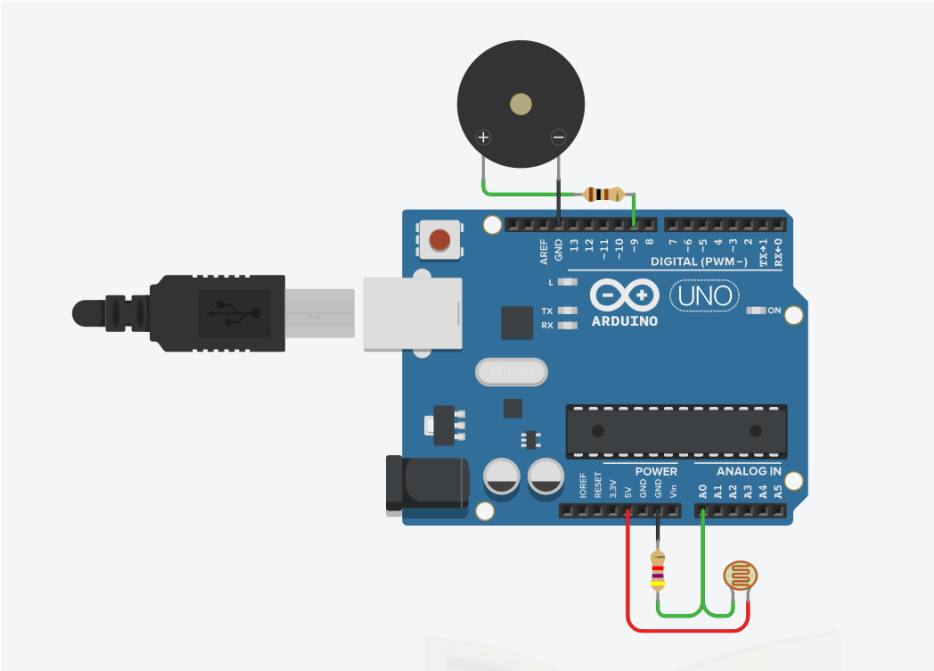
notes.arpanadhikari7.com.np

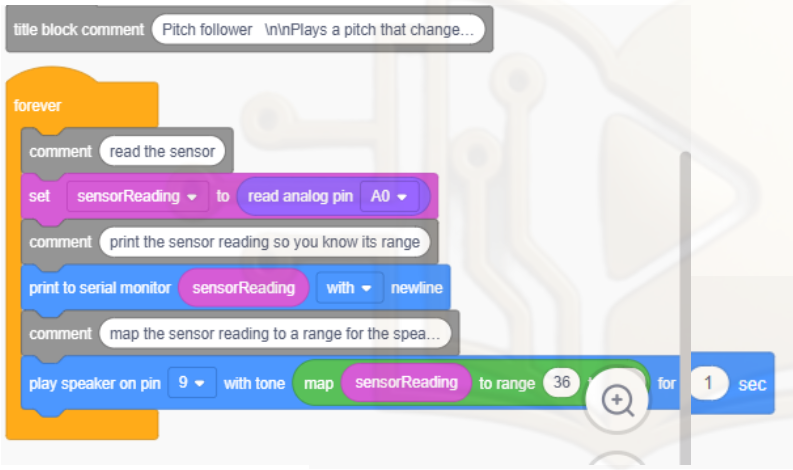




11. Tone Pitch follower

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12. Tone Keyword

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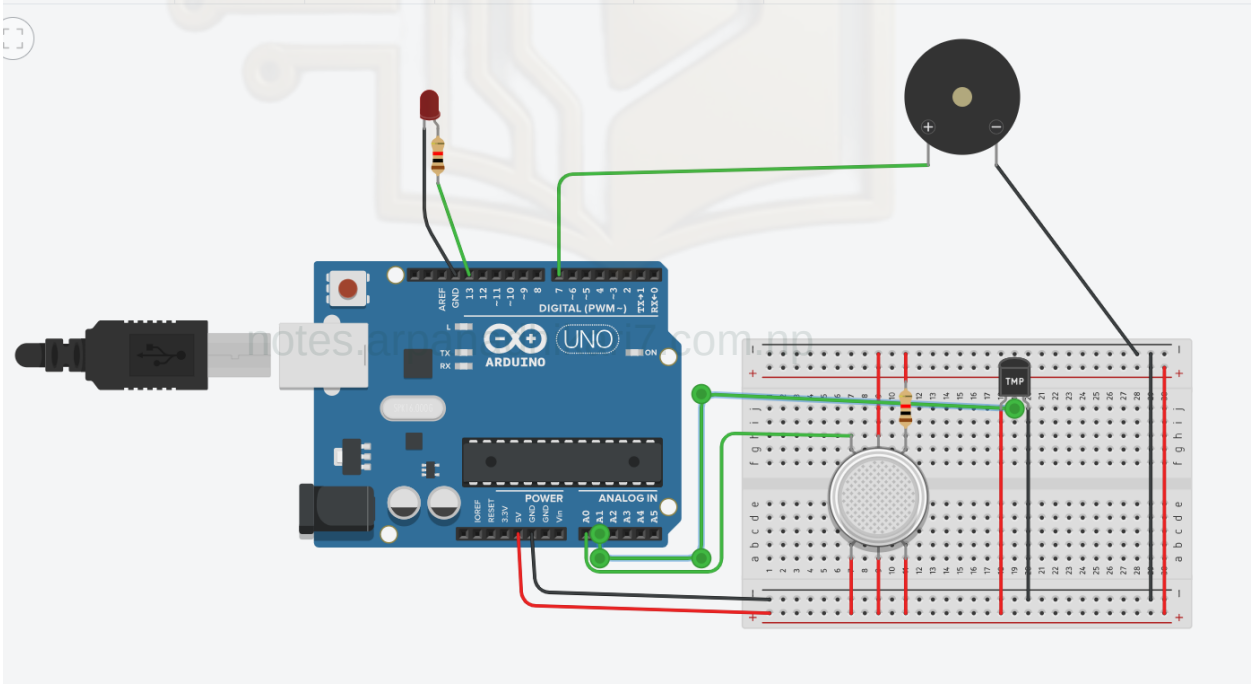
13. Interfacing Temperature Sensor with Arduino using Tinker CAD

```
// Fire Alarm System with Gas + Temperature + Buzzer + LED

const int gasPin = A0;        // MQ2 Smoke/Gas Sensor
const int tempPin = A1;       // Temperature sensor (LM35/TMP36)
const int buzzer = 7;         // Buzzer
const int led = 13;           // LED
int gasThreshold = 300;       // Gas threshold (adjust after testing)
float tempThreshold = 50.0;   // Celsius threshold for fire
void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(led, OUTPUT);
  Serial.begin(9600);
  Serial.println("Fire Alarm System Ready");
}
void loop() {
  // Read Gas
  int gasValue = analogRead(gasPin);
  // Read Temp
  int tempValue = analogRead(tempPin);
```

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```
 float voltage = tempValue * (5.0 / 1023.0);
 float tempC = (voltage - 0.5) * 100.0;  // TMP36 formula (if LM35 use voltage*100)
 // Serial output
 Serial.print("Gas Value = ");
 Serial.print(gasValue);
 Serial.print(" | Temp = ");
 Serial.print(tempC);
 Serial.println(" °C");
 // Alarm condition
 if (gasValue > gasThreshold || tempC > tempThreshold) {
   digitalWrite(led, HIGH);
   digitalWrite(buzzer, HIGH);
   delay(200);
   digitalWrite(buzzer, LOW);
   delay(200);
 }
 else {
   digitalWrite(led, LOW);
   digitalWrite(buzzer, LOW);
 }
 delay(500);
}
```

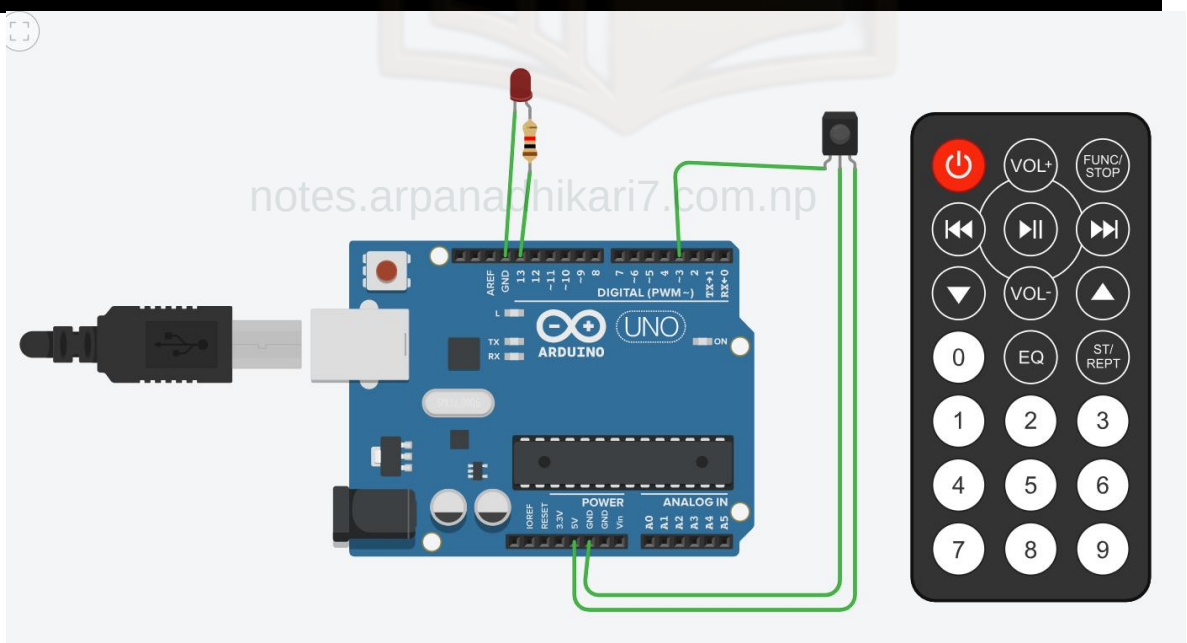


14. Controlling LEDs using an IR Remote

```
#include <IRremote.hpp>
const int rcvPin = 3;
```

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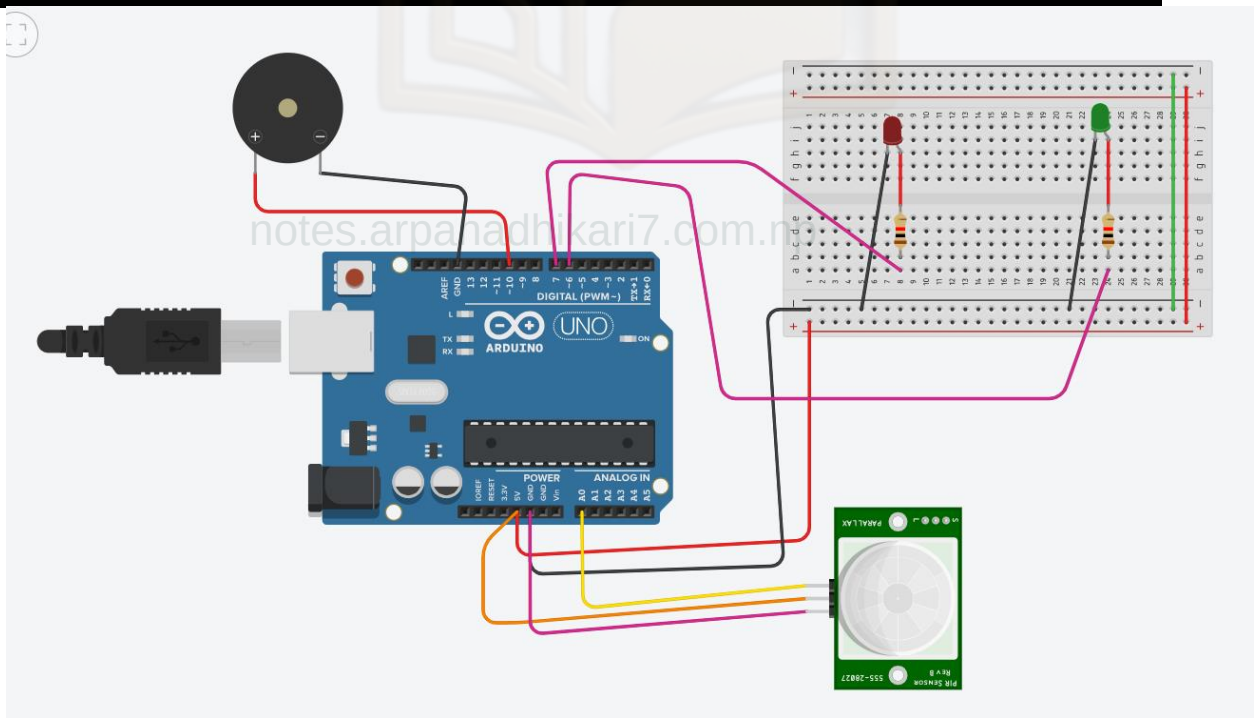
```
const int ledPin = 5;
void setup() {
  Serial.begin(9600);
  IrReceiver.begin(rcvPin, ENABLE_LED_FEEDBACK); // Correct initialization
  pinMode(ledPin, OUTPUT);
  Serial.println("IR Receiver Ready");
}
void loop() {
  if (IrReceiver.decode()) {
    unsigned long value = IrReceiver.decodedIRData.decodedRawData;
    Serial.println(value);
    switch (value) {
      case 4010852096:   // Button 1
        Serial.println("Button 1 Pressed");
        digitalWrite(ledPin, HIGH);
        break;
      case 3994140416:   // Button 2
        Serial.println("Button 2 Pressed");
        digitalWrite(ledPin, LOW);
        break;
      default:
        Serial.println("Unknown Button");
    }
    IrReceiver.resume(); // IMPORTANT
  }
}
```



15. PIR motion sensor

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```
int pirsensor = 0;
void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(10, OUTPUT);
}
void loop()
{
  pirsensor = analogRead(A0);
  Serial.println(pirsensor);
  if (pirsensor >= 100) {
    digitalWrite(7, LOW);
    digitalWrite(6, HIGH);
    tone(10, 92, 100); // play tone 30 (F#2 = 92 Hz)
  } else {
    digitalWrite(7, HIGH);
    digitalWrite(6, LOW);
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```



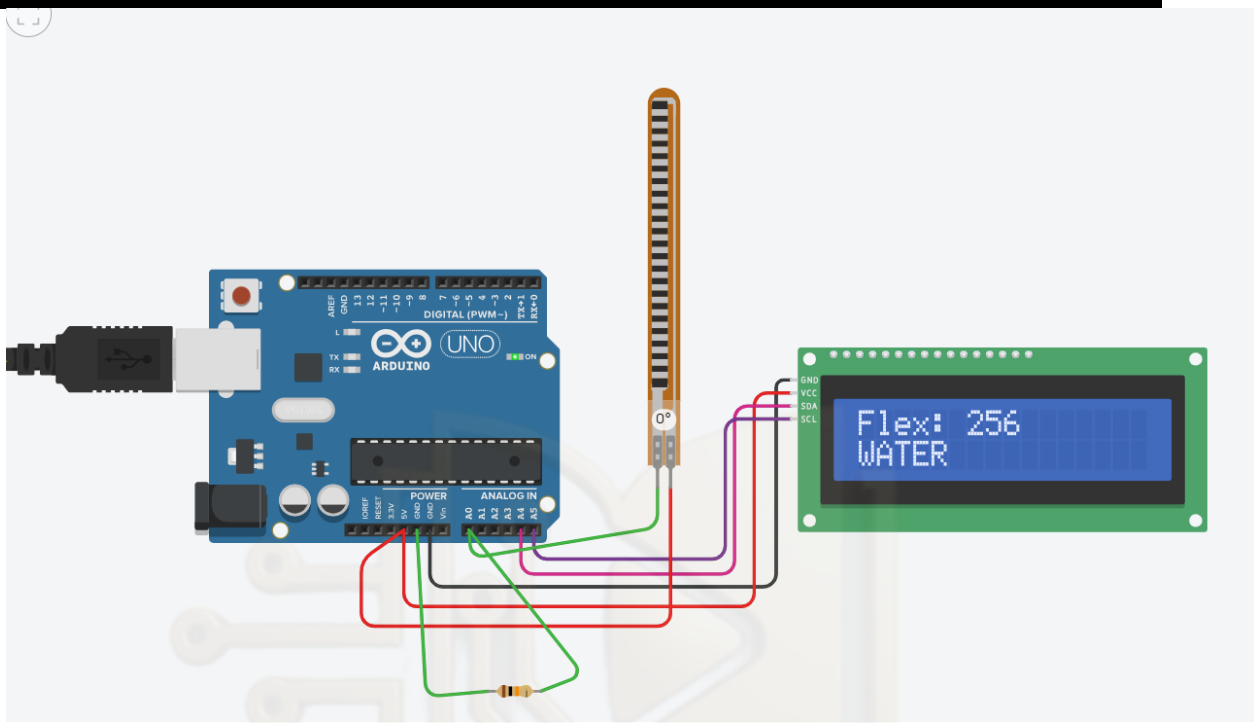
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16. Flex sensor for sign language

```
#include <Adafruit_LiquidCrystal.h>
Adafruit_LiquidCrystal lcd(0);
const int flexPin = A0;
int stableValue = 0;
void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  // Force reference stability
  analogReference(DEFAULT);
  // Warm up ADC
  for (int i = 0; i < 50; i++) {
    analogRead(flexPin);
    delay(10);
  }
  stableValue = analogRead(flexPin);
}
int readStableFlex() {
  long sum = 0;
  for (int i = 0; i < 20; i++) {
    sum += analogRead(flexPin);
    delay(5);
  }
  return sum / 20;
}
void loop() {
  int newValue = readStableFlex();
  // HARD deadband (prevents self-change)
  if (abs(newValue - stableValue) > 15) {
    stableValue = newValue;
  }
  Serial.print("Flex: ");
  Serial.println(stableValue);
  lcd.setCursor(0, 0);
  lcd.print("Flex: ");
  lcd.print(stableValue);
  lcd.print("    ");
  lcd.setCursor(0, 1);
  if (stableValue < 50) lcd.print("NO SENSOR     ");
  else if (stableValue < 200) lcd.print("HAND OPEN     ");
  else if (stableValue < 400) lcd.print("WATER         ");
```

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```
  else if (stableValue < 600) lcd.print("FOOD         ");
  else if (stableValue < 800) lcd.print("BATHROOM      ");
  else lcd.print("HELP         ");
  delay(200);
}
```



Do Yourself

17. **Password-Protected Security System project using Arduino, Keypad, Piezo Buzzer, and LED**
18. **Voting system with LCD**
19. **Light Intensity Measurement using LDR sensor**
20. **Vibration sensor**

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